



Community Project
Branch: CSE-CYBERSECURITY & INTERNET OF THINGS(ICB)
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- **Project Title: VIDYA Uday – DIGITAL EMPOWERMENT AND AI AWARENESS**
- **Batch Number:3**

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Abstract

A concise overview of VidyaUday and its core purpose



Background

- Rural 10th and Intermediate students in Andhra Pradesh and Telangana face a severe guidance deficit, remaining largely unaware of higher education pathways, statutory government scholarships, and free AI learning tools.
- Essential academic information remains fragmented across dense government gazettes, English-dominant portals, and aggressive commercial coaching apps charging predatory fees.

Our Solution

- VidyaUday provides a 100% free, mobile-optimized, bilingual web platform requiring zero login, no app installation, and zero personal data collection.
- Integrates dual-state scholarship eligibility matching, interactive career roadmaps (MPC, BiPC, CEC, Polytechnic, ITI), SCERT digital textbooks, and a dedicated Digital Literacy & Cyber Scam Buster Hub.

Expected Outcome

- Eliminates dependency on expensive cyber cafés, maximizes statutory scholarship claim rates across rural districts, and empowers first-generation learners with confident digital safety.

Introduction

The challenge rural students face and what VidyaUday does about it



The Growing Rural Challenge

- Over 12 lakh students complete 10th and Intermediate board exams annually across Andhra Pradesh and Telangana, yet rural students lack structured career mentoring and make life-defining decisions blindly.
- Commercial coaching institutes exploit this vacuum by charging exorbitant fees (₹25,000–₹50,000), leaving economically disadvantaged rural students behind.

The Information & Language Barrier

- While excellent government welfare schemes and digital resources exist, they are scattered across fragmented portals in complex English without vernacular guidance.
- Rural students frequently miss application deadlines for fully funded government scholarships due to procedural obscurity and lack of digital awareness.

Our Platform: VidyaUday

- VidyaUday bridges this digital divide as a lightweight, bilingual (English & Telugu) portal uniting career paths, scholarship finders, AI study prompts, and cyber safety tools under one roof.

Problem Statement

Rural students are not short on ability — they are short on access



Core Problem Statement

Lakhs of rural 10th and Intermediate students in Andhra Pradesh and Telangana face a severe guidance deficit, making critical educational and career decisions blindly without awareness of statutory government scholarships, digital learning platforms, or modern technology tools. Fragmented government notifications, dense English-only web portals, and aggressive commercial coaching traps systematically disenfranchise first-generation rural learners.

Key Pain Points Identified in Rural Field Survey

- Information Asymmetry: 74.2% of surveyed rural students are unaware of viable career alternatives beyond traditional MPC/BiPC streams.
- Unclaimed Welfare Aid: Over ₹50,000 per student in statutory government scholarships remains unclaimed annually due to lack of deadline visibility.
- The Digital Divide: 81.5% of students have never utilized free educational AI tools or cyber hygiene practices for academic advancement.

Project Objectives

Five clear goals that define what VidyaUday sets out to achieve



Strategic Project Objectives

- **Bilingual Career Guidance:** Provide structured, year-by-year career path navigators for 10th and Intermediate streams (MPC, BiPC, CEC, MEC, Polytechnic, ITI) in English and Telugu.
- **Centralized Scholarship Discovery:** Streamline statutory Central and State schemes (AP JVD, TS ePASS, NMMS, Pragati) with instant dual-state eligibility matching and deadline tracking.
- **Democratize Educational AI:** Curate high-impact, 1-click copyable AI study prompts enabling rural students to leverage Gemini and ChatGPT for mathematics, essays, and vernacular explanations.
- **Cyber Literacy & Scam Prevention:** Equip rural first-generation digital users with cyber safety rules, digital readiness self-assessments, and official 1930 cybercrime reporting guidance.
- **Zero-Barrier Universal Access:** Deliver an ultra-lightweight, ad-free web platform that operates smoothly on low-cost smartphones and 3G networks without requiring registration or logins.

Project Scope

What VidyaUday includes — and what it deliberately does not cover



In-Scope (Platform Capabilities)

- **Target Audience:** Comprehensive coverage tailored specifically for rural 10th (SSC), Intermediate (+2), Diploma/Polytechnic, and ITI students across Andhra Pradesh and Telangana.
- **Functional Modules:** Integrated Career Roadmap Explorer, AP/TS Dual Eligibility Scholarship Matcher, Scheme Comparison Matrix, SCERT Textbook Hub, AI Prompts, and Digital Scam Buster.
- **Accessibility & Reach:** 100% responsive design optimized for entry-level mobile devices, offline study pack downloads, and bilingual English-Telugu synthesized voice assistance.

Out-of-Scope (Deliberate Design Boundaries)

- **Not a Classroom Replacement:** Designed strictly as an informational and career guidance bridge, not as a substitute for classroom teachers, curricula, or formal school instruction.
- **No Direct Financial Transactions:** Does not disburse scholarship funds or collect banking credentials; directly redirects students to official government portals (NSP, ePASS).
- **Zero Backend Data Harvesting:** Operates purely client-side without storing user personal data, phone numbers, or passwords, ensuring complete student privacy.



Key Research Findings & Literature Analysis

- National Education Policy (NEP 2020) Recommendations: Emphasizes early career mentoring, mother-tongue educational resources, and integration of digital literacy into secondary school stages.
- ASER & Field Research on Rural EdTech: ASER reports indicate that while smartphone access in rural households exceeds 75%, educational use is heavily hindered by English-only interfaces and complex navigation.
- Evaluation of Official Government Portals: Portals such as NSP and State ePASS provide essential funding but feature fragmented multi-layered user interfaces that require cyber café mediation.
- Commercial EdTech Platform Barriers: Commercial platforms lock guidance behind paywalls of ₹20,000–₹50,000/year, catering almost exclusively to urban English-medium students.
- Synthesis & Need for VidyaUday: Highlights an urgent societal need for a unified, bilingual, lightweight portal offering centralized, free guidance directly to students without intermediaries.

VidyaUday vs Competitors

Feature-by-feature comparison — what makes VidyaUday unique



Feature / Capability	VidyaUday	Govt Portals (NSP)	Private EdTech	Static PDFs
Access Cost & Subscription	100% Free Forever	Free (Complex UI)	₹20,000–₹50,000	Free
Language Support	English & Telugu (Full)	English / Regional Mix	Predominantly English	Single Language
Interactive Career Roadmaps	Yes (Year-by-Year Paths)	No (Raw Notices)	Yes (Paid Paywall)	Static Lists
Dual AP/TS Scholarship Checker	Yes (30-Sec Quiz & Matrix)	Separate State Logins	No (Commercial Only)	No
Curated Student AI Prompts	Yes (1-Click Copy Prompts)	No	Paid AI Tutors	No
Digital Literacy & Scam Buster	Yes (Quiz + Scam Guide)	No	No	No

Data Gathered / Data Used

Every module is powered by real, verified field and government data



Primary Field Research (MVGR CSP Outreach Survey)

- Empirical Student Survey: Conducted field surveys and interactive sessions across 120+ rural high school and intermediate students in Vizianagaram district and adjoining mandals.
- Career Awareness Findings: 74.2% of surveyed students were unaware of alternative diploma, vocational, or allied health options beyond MPC/BiPC.
- Technology Gap: 81.5% of students had never used free AI assistance tools for homework, math problem-solving, or vocabulary enhancement.
- Scholarship Disconnect: 68.4% of eligible students missed statutory financial aid deadlines due to procedural complexity and lack of notification.

Secondary Data & Statutory Government Gazette Sources

- Government Welfare Orders: Formulated scholarship rule engines using official AP Social Welfare Department Gazettes (G.O. Ms. No. 77 & 115) for Jagananna Vidya Deevena & Vasathi Deevena.
- State & Central Schemes: Codified eligibility rules from Telangana ePASS (CGG) and National Scholarship Portal (NSP Post-Matric, NMMS, Pragati, Saksham).
- Official Curriculum Data: Mapped authentic digitized textbook repositories from AP SCERT and TS SCERT official academic portals.

Methodology / System Design — Flow Chart

A guided five-step journey from confusion to a confident, informed decision



1. Student Access

Visits lightweight, zero-login web platform on mobile/desktop; chooses English or Telugu.

2. Module Selection

Selects Career Roadmaps, Scholarship Finder, SCERT Books, AI Hub, or Scam Buster.

3. Interactive Input

Inputs simple criteria: class level, stream of interest, family income, and state (AP/TS).

4. Logic Execution

Client-side JS rule engine evaluates eligibility matrix, filters schemes, and renders results.

5. Empowered Action

Student accesses official application links, downloads offline guides, and applies safely.

Methodology / System Design — Block Diagram

Four-layer architecture that powers the VidyaUday platform



Layer 1: User Interface & Presentation Layer

Responsive Mobile-First UI (HTML5/CSS3) | Bilingual Switcher (English/Telugu) | Web Speech Voice Assistant | High-Contrast Typography

Layer 2: Core Feature & Application Layer

Interactive Career Roadmaps | AP/TS Dual Scholarship Matcher | Scheme Comparison Matrix | SCERT Books Hub | AI Prompts | Scam Buster

Layer 3: Client-Side Logic & Offline Storage

JavaScript Dynamic Filter Engines | Eligibility Decision Tree Matrix | Browser LocalStorage | Offline Resource Pack Generator

Layer 4: External Data & Verification Layer

Official AP Social Welfare (JVD) Rules | Telangana ePASS (CGG) Data | National Scholarship Portal (NSP) | AP/TS SCERT Curriculum

Implementation / Core Modules

Six comprehensive modules built and verified on the live platform

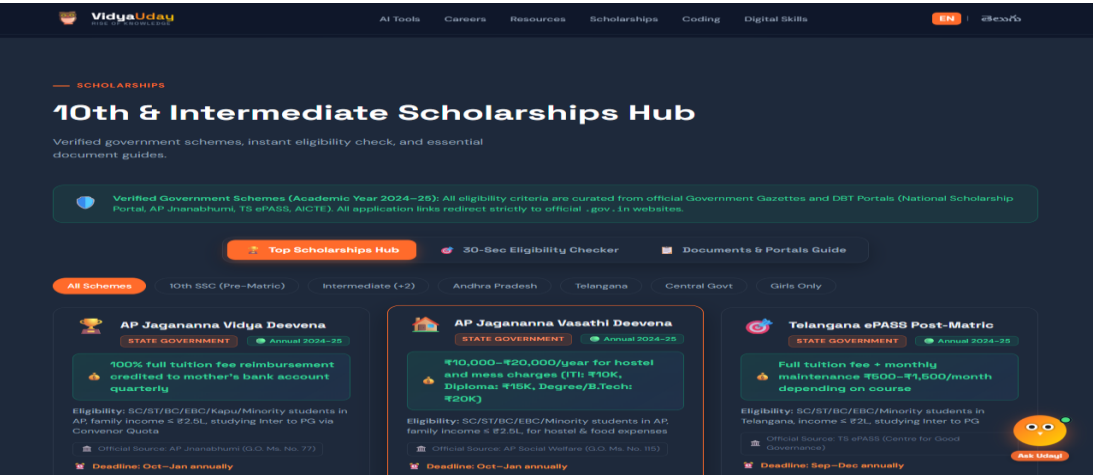


Platform Implementation Modules

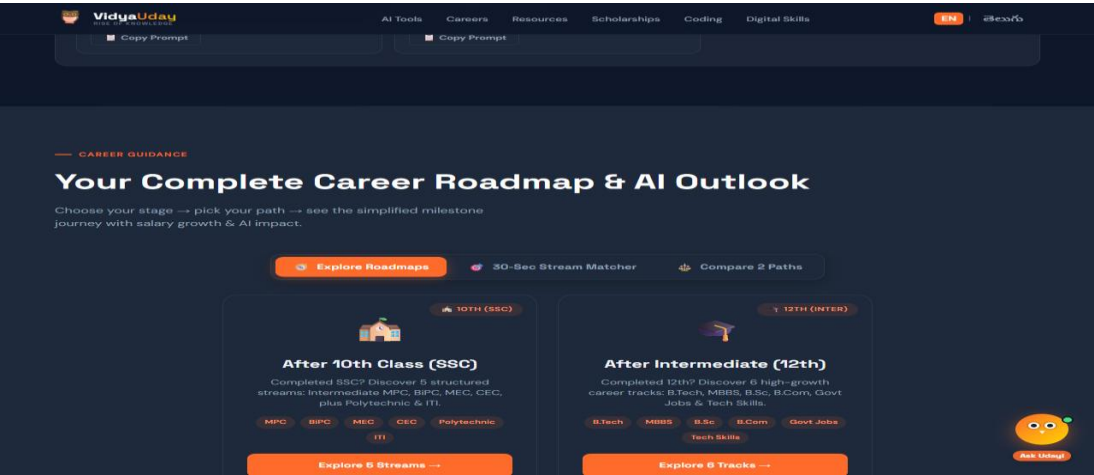
- 1. Interactive Career Roadmap Explorer: Year-by-year academic and career path mapping after 10th and 12th covering MPC, BiPC, CEC, MEC, Polytechnic, and ITI with degree outcomes and entrance exams.
- 2. AP/TS Dual Scholarship Eligibility Engine: Interactive 30-second checker evaluating state, caste, annual income, and class level against AP JVD, TS ePASS, NSP Post-Matric, NMMS, and AICTE Pragati schemes.
- 3. Scheme Comparison & Application Matrix: Side-by-side comparative table analyzing benefits, eligibility caps, deadline calendars, and direct verified official application portals.
- 4. Curated Student AI Study Prompts: Categorized, 1-click copyable prompts for Google Gemini and ChatGPT assisting with homework, math solutions, essay outlining, and English-Telugu translation.
- 5. Digital Literacy & Cyber Scam Buster: 3-mode hub featuring student digital safety checklists, a 30-second readiness self-assessment quiz, and practical guides to detect phishing and fake scholarship traps.
- 6. Universal Accessibility & Offline Packs: 1-click downloadable printable study and scholarship guides, coupled with bilingual speech synthesis for vernacular auditory learners.

VidyaUday Live Platform Screenshots

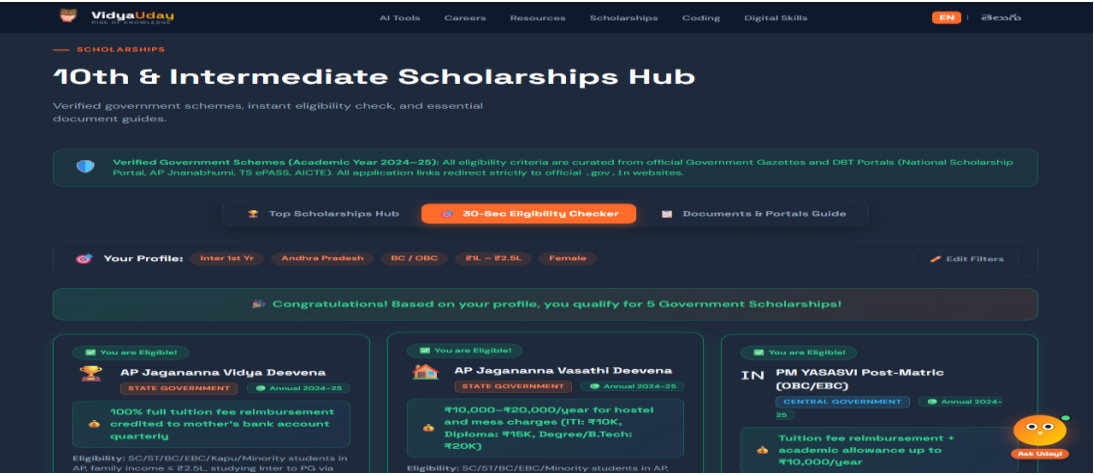
Real production screens demonstrating interactive student capabilities



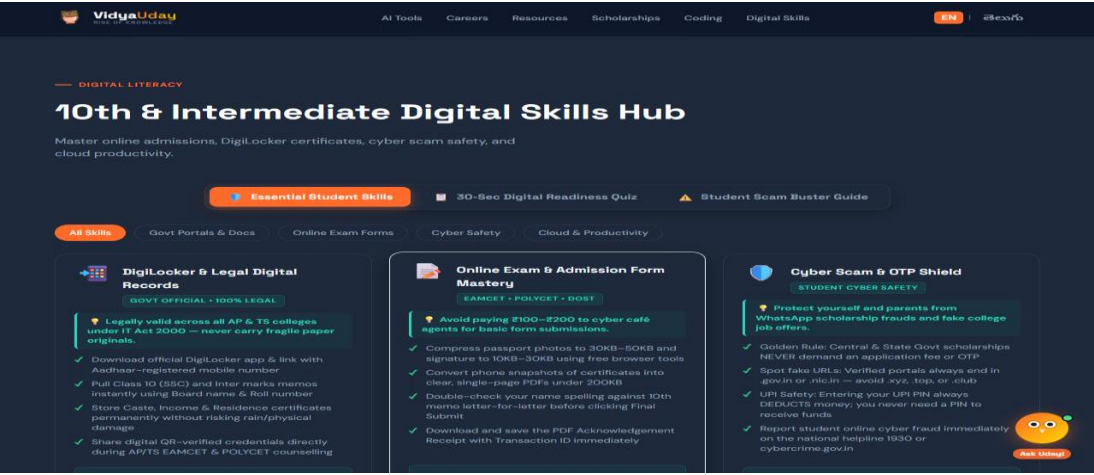
Screen 1: Scholarship Discovery Hub



Screen 2: Interactive Career Roadmaps



Screen 3: AP/TS Eligibility Matcher



Screen 4: Digital Literacy & Scam Buster



Social Impact on Rural Student Communities

- **Democratized Career Guidance:** Equips rural students with structured, multi-stream career visibility, breaking generational cycles of choosing courses purely based on local peer pressure.
- **Elimination of Exploitative Intermediaries:** Empowers students and parents to understand scholarship eligibility and application procedures directly, eliminating cyber café commission fees (₹200–₹500/application).
- **Digital Empowerment & Cyber Safety:** Promotes responsible online hygiene, digital self-confidence, and proactive scam defense among vulnerable first-generation internet users.

Technological Impact & Innovation

- **Zero-Barrier Universal Access:** High-speed, responsive performance that loads in under 1.5 seconds on 3G/4G rural networks with zero battery or data overhead.
- **Privacy-First Architecture:** 100% client-side data evaluation with zero database storage, ensuring complete compliance with student digital privacy standards.

Challenges Faced

Key obstacles encountered during development and how each one was resolved



Key Challenges & Engineering Solutions

- **State-Specific Rule Discrepancies:** Challenge: AP and Telangana maintain divergent scholarship eligibility criteria, caste classifications, and portal architectures. Solution: Engineered a unified, state-aware JSON decision matrix that dynamically maps student inputs to state-specific statutory guidelines.
- **Rural Bandwidth & Device Constraints:** Challenge: Target rural students access the web on low-spec smartphones over unstable 3G/4G mobile connections. Solution: Replaced resource-heavy frameworks with pure lightweight JavaScript, CSS grid layouts, SVG icons, and printable offline download packs.
- **Vernacular Language & Terminology Translation:** Challenge: Complex government circulars and academic guidance are unintelligible to rural students in English. Solution: Authored authentic bilingual Telugu translations paired with browser-native Web Speech API voice synthesis for auditory accessibility.

Conclusion

What VidyaUday proves — and what it means for every rural student in AP & TS



Summary of Project Achievements

- Successfully engineered and deployed VidyaUday as a 100% free, production-ready community educational empowerment platform.
- Centralized diverse educational tracks, statutory AP/TS scholarships, digitized SCERT textbooks, and student AI prompts into a unified single-page portal.
- Integrated practical digital literacy drills and cyber scam defense guides to protect students from predatory online coaching and scholarship fraud.

Final Project Takeaway

- VidyaUday demonstrates that accessible, thoughtful technology can bridge the urban-rural educational divide. By placing high-quality guidance, scholarship access, and digital literacy directly into students' hands at zero cost, we empower rural youth to make confident, informed choices for their future.



Roadmap for Future Platform Enhancements

- Progressive Web App (PWA) Offline Engine: Implement complete service worker caching to allow full offline operation on village smartphones even when network connectivity drops completely.
- Automated WhatsApp / Telegram Community Bot: Develop an automated vernacular bot sending automated scholarship deadline reminders, entrance exam alerts, and career FAQs directly to students' family phones.
- College Student Mentorship Network: Establish a digital volunteer mentorship bridge connecting senior engineering students from MVGR College with rural school students for 1-on-1 career guidance.
- Native Ultra-Lightweight Android APK: Package the platform into a standalone 2MB Android application distributed freely through village schools, Grama Sachivalayams, and local libraries.



Academic & Statutory References

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